Question 1

Install a new software package using Kali Linux repositories

Approach

I used Kali Linux package management tools to install software from the official repository. First, I updated the package list and then installed a package using the apt package manager.

Commands

- `sudo apt update` → updates package list from repositories
- `sudo apt install <package-name>` → installs the selected software package
- `apt list --installed` → shows installed packages

Output

STEP 1 : Update repository package list

```
┌──(aman㉿kali)-[~]
└─$ sudo apt update
[sudo] password for aman:
Hit:1 http://http.kali.org/kali kali-rolling InRelease
434 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

STEP 2 : Install a software package

```
┌──(aman㉿kali)-[~]
└─$ sudo apt install nmap
nmap is already the newest version (7.98+dfsg-1kali1).
The following packages were automatically installed and are no longer required:
  curlftpfs      libavformat61          libfuse2t64            libmpeg2encpp-2.1-0t64  libplacebo351     librubberband2  libsphinxbase3t64  mesa-vdpau-drivers  python3-aiomcache       python3-wapiti-arsenic  vdpau-driver-all
  libaudio2      libconfig-inifiles-perl libgav1-1             libmplex2-2.1-0t64      libpocketsphinx3  libsframe2      libswscale8        pocketsphinx-en-us  python3-fs              python3-yaswfp
  libavfilter10  libcrypt-dev           libmjpegutils-2.1-0t64 libmupdf25.1            libpostproc58     libsnmp40t64    libvdpau-va-gl1    python3-aiocache    python3-terminaltables  ruby-unf-ext
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 434
```

STEP 3 : Verify installation

```
┌──(aman㉿kali)-[~]
└─$ nmap --version
Nmap version 7.98SVN ( https://nmap.org )
Platform: x86_64-unknown-linux-gnu
Compiled with: nmap-liblua-5.4.8 openssl-3.5.5 nmap-libssh2-1.11.1 libz-1.3.1 nmap-libpcre2-10.47 nmap-libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select
```

Key Observations

- Kali Linux uses repositories to manage software packages
- `apt` automatically installs dependencies required for the software.
- Root privileges are required to install system packages.

What I learned

Kali Linux uses the APT package manager to install, update, and manage software efficiently from trusted repositories.

Question 2

Remove an installed software package from the system

Approach

After installing a package, I removed it using the APT remove command to understand how software uninstallation works in Linux.

Commands

- `sudo apt remove <package-name>` → removes installed package
- `sudo apt purge <package-name>` → removes package with configuration files
- `sudo apt autoremove` → removes unnecessary dependencies

Output

STEP 1 : Remove installed package

```
┌──(aman㉿kali)-[~]
└─$ sudo apt remove nmap
[sudo] password for aman:
The following packages were automatically installed and are no longer required:
  binutils-mingw-w64-base          gcc-mingw-w64-x86-64-win32-runtime  liblinear4          libswscale8              python3-aiomcache       python3-jeepney                  python3-pynfsclient     python3-winacl
  binutils-mingw-w64-i686          imagemagick                         libluajit-5.1-2     libvdpau-va-gl1          python3-aiosmb          python3-lsassy                   python3-pypsrp          python3-xmltodict
  binutils-mingw-w64-x86-64        imagemagick-7.q16                   libluajit-5.1-common medusa                  python3-aiowinreg       python3-masky                    python3-pypykatz        python3-yaswfp
  bloodhound.py                    krb5-multidev                       libmjpegutils-2.1-0t64 mesa-vdpau-drivers    python3-arc4            python3-minidump                 python3-pyshodan        rsh-redone-client
  comerr-dev                       libaudio2                           libmpeg2encpp-2.1-0t64 mingw-w64-common      python3-asciitree       python3-minikerberos             python3-pyspnego        ruby-unf-ext
  curlftpfs                        libavfilter10                       libmplex2-2.1-0t64  mingw-w64-i686-dev       python3-asn1tools       python3-msldap                   python3-pywerview       smtp-user-enum
  dnsmap                           libavformat61                       libmupdf25.1        mingw-w64-x86-64-dev     python3-asyauth         python3-neo4j                    python3-qasync          sparta-scripts
  dsniff                           libconfig-inifiles-perl             libnids1.21t64      ndiff                    python3-asysocks        python3-neobolt                  python3-qrcode          toilet-fonts
  ettercap-common                  libcrypt-dev                        libplacebo351       nmap-common              python3-bitstruct       python3-neotime                  python3-serial-asyncio  unicornscan
  ettercap-graphical               libfuse2t64                         libpocketsphinx3    oracle-instantclient-basic python3-browser-cookie3 python3-oscrypto                 python3-smmap           urlscan
  figlet                           libgav1-1                           libpostproc58       pocketsphinx-en-us       python3-dploot          python3-pefile                   python3-terminaltables  vdpau-driver-all
  finger                           libgssrpc4t64                       librubberband2      python3-aardwolf         python3-dunamai         python3-poetry-core              python3-tld             wapiti
  gcc-mingw-w64-base               libkadm5clnt-mit12                  libsframe2          python3-aesedb           python3-fs              python3-poetry-dynamic-versioning python3-tomlkit
  gcc-mingw-w64-i686-win32         libkadm5srv-mit12                   libsmb2-6           python3-aiocache         python3-git             python3-pyexploitdb              python3-unicrypto
  gcc-mingw-w64-i686-win32-runtime libkdb5-10t64                       libsnmp40t64        python3-aiocmd           python3-gitdb           python3-pyfiglet                 python3-wapiti-arsenic
  gcc-mingw-w64-x86-64-win32       libkrb5-dev                         libsphinxbase3t64   python3-aioconsole       python3-httpx-ntlm      python3-pylnk3                   python3-wapiti-swagger
Use 'sudo apt autoremove' to remove them.

REMOVING:
  commix  kali-linux-default  kali-linux-headless  kali-tools-top10  legion  metasploit-framework  msfpc  netexec  nmap  python3-libnmap  set  unicorn-magic  zenmap

Summary:
  Upgrading: 0, Installing: 0, Removing: 13, Not Upgrading: 430
  Freed space: 638 MB

Continue? [Y/n] y
(Reading database… 438670 files and directories currently installed.)
Removing kali-linux-default (2026.1.7)…
Removing kali-linux-headless (2026.1.7)…
Removing commix (4.1-0kali1)…
Removing kali-tools-top10 (2026.1.7)…
Removing legion (0.5.2-0kali1)…
Removing unicorn-magic (3.12-0kali3)…
Removing set (8.0.3+git20241021-0kali1)…
Removing msfpc (1.4.5-0kali3)…
Removing netexec (1.5.1-0kali2)…
Removing zenmap (7.98+dfsg-1kali1)…
Removing python3-libnmap (0.7.3-1)…
Removing metasploit-framework (6.4.112-0kali1)…
Removing nmap (7.98+dfsg-1kali1)…
Processing triggers for desktop-file-utils (0.28-1)…
Processing triggers for hicolor-icon-theme (0.18-2)…
Processing triggers for libc-bin (2.42-13)…
Processing triggers for man-db (2.13.1-1)…
Processing triggers for wordlists (2026.1.2)…
Processing triggers for kali-menu (2026.1.3)…
```

STEP 2 : Remove unused dependencies

```
┌──(aman㉿kali)-[~]
└─$ sudo apt autoremove
REMOVING:
  binutils-mingw-w64-base          gcc-mingw-w64-x86-64-win32-runtime  liblinear4          libswscale8          python3-aiomcache    python3-jeepney          python3-pynfsclient  python3-winacl
  binutils-mingw-w64-i686          imagemagick                         libluajit-5.1-2     libvdpau-va-gl1      python3-aiosmb       python3-lsassy           python3-pypsrp       python3-xaltodict
  binutils-mingw-w64-x86-64        imagemagick-7.q16                   libluajit-5.1-common medusa              python3-aiowinreg    python3-masky            python3-pypykatz     python3-yaswfp
  bloodhound.py                    krb5-multidev                       libmjpegutils-2.1-0t64 mesa-vdpau-drivers python3-arc4        python3-minidump         python3-pyshodan     rsh-redone-client
  comerr-dev                       libaudio2                           libmpeg2encpp-2.1-0t64 mingw-w64-common  python3-asciitree    python3-minikerberos     python3-pyspnego     ruby-unf-ext
  curlftpfs                        libavfilter10                       libmplex2-2.1-0t64  mingw-w64-i686-dev   python3-asn1tools    python3-msldap           python3-pywerview    smtp-user-enum
  dnsmap                           libavformat61                       libmupdf25.1        mingw-w64-x86-64-dev python3-asyauth      python3-neo4j            python3-qasync       sparta-scripts
  dsniff                           libconfig-inifiles-perl             libnids1.21t64      ndiff                python3-asysocks     python3-neobolt          python3-qrcode       toilet-fonts
  ettercap-common                  libcrypt-dev                        libplacebo351       nmap-common          python3-bitstruct    python3-neotime          python3-serial-asyncio unicornscan
  ettercap-graphical               libfuse2t64                         libpocketsphinx3    oracle-instantclient-basic python3-browser-cookie3 python3-oscrypto      python3-smmap        urlscan
  figlet                           libgav1-1                           libpostproc58       pocketsphinx-en-us   python3-dploot       python3-pefile           python3-terminaltables vdpau-driver-all
  finger                           libgssrpc4t64                       librubberband2      python3-aardwolf     python3-dunamai      python3-poetry-core      python3-tld          wapiti
  gcc-mingw-w64-base               libkadm5clnt-mit12                  libsframe2          python3-aesedb       python3-fs           python3-poetry-dynamic-versioning python3-tomlkit
  gcc-mingw-w64-i686-win32         libkadm5srv-mit12                   libsmb2-6           python3-aiocache     python3-git          python3-pyexploitdb      python3-unicrypto
  gcc-mingw-w64-i686-win32-runtime libkdb5-10t64                       libsnmp40t64        python3-aiocmd       python3-gitdb        python3-pyfiglet         python3-wapiti-arsenic
  gcc-mingw-w64-x86-64-win32       libkrb5-dev                         libsphinxbase3t64   python3-aioconsole   python3-httpx-ntlm   python3-pylnk3           python3-wapiti-swagger

Summary:
  Upgrading: 0, Installing: 0, Removing: 124, Not Upgrading: 427
  Freed space: 1,296 MB

Continue? [Y/n] y
(Reading database… 405929 files and directories currently installed.)
Removing gcc-mingw-w64-i686-win32 (15.2.0-12+28.3)…
Removing binutils-mingw-w64-i686 (2.45.90.20260125-1+13.3)…
Removing gcc-mingw-w64-x86-64-win32 (15.2.0-12+28.3)…
Removing binutils-mingw-w64-x86-64 (2.45.90.20260125-1+13.3)…
Removing binutils-mingw-w64-base (2.45.90.20260125-1+13.3)…
Removing bloodhound.py (1.9.0-0kali1)…
Removing libkrb5-dev:amd64 (1.22.1-2)…
Removing krb5-multidev:amd64 (1.22.1-2)…
Removing comerr-dev:amd64 (2.1-1.47.2-3+b8)…
Removing curlftpfs (0.9.2-10)…
Removing dnsmap (0.36-4)…
Removing dsniff (2.5a2-1)…
Removing ettercap-graphical (1:0.8.4-5)…
Removing ettercap-common (1:0.8.4-5)…
Removing figlet (2.2.5-3.1)…
Removing finger (0.17-17)…
Removing gcc-mingw-w64-x86-64-win32-runtime (15.2.0-12+28.3)…
Removing gcc-mingw-w64-i686-win32-runtime (15.2.0-12+28.3)…
```

Key Observations

- Removing packages helps maintain system performance.
- `autoremove` cleans unnecessary dependencies.
- `purge` removes configuration files completely.

What I learned

Linux allows full control over installed software and unused dependencies can be cleaned to optimize the system.

Question 3

Update the system repository information

Approach

I refreshed the system's package database to ensure I have the latest information about available software versions.

Commands

- `sudo apt update` → refreshes repository package index

Output

STEP 1 : Update repository information

```
┌──(aman㉿kali)-[~]
└─$ sudo apt update
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.0 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.5 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [119 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [276 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [186 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [892 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [14.3 kB]
Fetched 76.0 MB in 20s (3,801 kB/s)
489 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Key Observation

- Repository update does not install software.
- It synchronizes package metadata with online servers.

What I learned

keeping repositories updated ensures access to latest software versions and security patches.

Question 4

Upgrade installed software packages to latest versions

Approach

After updating repository information, I upgraded installed software packages to the latest available versions.

Commands

- `sudo apt upgrade` → upgrades installed packages
- `sudo apt full-upgrade` → upgrades packages with dependency changes

Output

STEP 1 : Upgrade installed packages

```
┌──(aman㉿kali)-[~]
└─$ sudo apt upgrade
[sudo] password for aman:
The following packages were automatically installed and are no longer required:
  libdlt2  libdnnl3.6  libgdk-pixbuf2.0-bin  libonnx1t64  libonnxruntime-providers  libonnxruntime1.21  libtsk19t64  libtss2-fapi1t64  libxnnpack0.20241108  libzxing3
Use 'sudo apt autoremove' to remove them.

Upgrading:
  7zip                      git-man                   libblockdev-smart3        libicu78                  libsyn123-0t64            openssh-server            python3-talloc
  acl                       gnome-keyring             libblockdev-swap3         libip4tc2                 libsz2                    openssh-sftp-server       python3-tdb
  adwaita-icon-theme        gnome-keyring-pkcs11      libblockdev-utils3        libip6tc2                 libtag-c2                 openvas-scanner           python3-tinycss2
  apache2                   gnupg                     libblockdev3              libitm1                   libtag2                   orca                      python3-tldextract
  apache2-bin               gnupg-l10n                libbluray3                libjavascriptcoregtk-4.1-0 libtalloc2               pci.ids                   python3-tornado
  apache2-data              gnupg-utils               libbpf1                   libjson-c5                libtdb1                   pdf-parser                python3-tz
  apache2-utils             gnutls-bin                libbrlapi0.8              libksba8                  libtevent0t64             pinentry-curses           python3-uc-micro
  apt                       gpg                       libbson2-2                libldb2                   libtimedate-perl          pinentry-gnome3           python3-urllib3
  apt-utils                 gpg-agent                 libcap-ng0                liblerc4                  libtsan2                  pipx                      python3-urwid
  autoconf                  gpg-wks-client            libcap2                   libllv-0-0                libtss2-esys-3.0.2-0t64   postgresql-18             python3-virtualenv
  automake                  gpgconf                   libcap2-bin               libllvmspirvlib18.1       libtss2-fapi1t64          postgresql-18-jit         python3-webcolors
  bind9-dnsutils            gpgsm                     libcc1-0                  liblouis-data             libtss2-mu-4.0.1-0t64     postgresql-18-pg-gvm      python3.14-tk
  bind9-host                gpgv                      libclone-perl             liblouis20                libtss2-sys1t64           postgresql-client-18      rake
  bind9-libs                greenbone-feed-sync       libcompress-raw-lzma-perl liblsan0                  libtss2-tcti-cmd0t64      proj-bin                  ristretto
  binutils                  gstreamer1.0-gl           libctf-nobfd0             liblua5.3-0               libtss2-tcti-device0t64   proj-data                 ruby-addressable
  binutils-common           gstreamer1.0-libav        libctf0                   liblwp-protocol-https-perl libtss2-tcti-libtpms0t64 publicsuffix              ruby-cms-scanner
  binutils-x86-64-linux-gnu gstreamer1.0-plugins-bad  libcurl3t64-gnutls       libmagickcore-7.q16-10    libtss2-tcti-mssim0t64    pv                        ruby-connection-pool
  binwalk3                  gstreamer1.0-plugins-base libcurl4t64               libmagickcore-7.q16-10-extra libtss2-tcti-spi-helper0t64 python-matplotlib-data ruby-ethon
  burpsuite                 gstreamer1.0-plugins-good libdate-manip-perl        libmagickwand-7.q16-10    libtss2-tcti-swtpm0t64    python-tinycss2-common    ruby-json
  busybox                   gstreamer1.0-x            libdbus-1-3               libmariadb3               libtss2-tctildr0t64       python3-anyio             ruby-logging
  ca-certificates-java      guymager                  libdw1t64                 libmongoc2-2              libubsan1                 python3-astroid           ruby-mime
  chntpw                    gvfs                      libelf1t64                libmosquitto1             libudisks2-0              python3-async-timeout     ruby-mime-types-data
  chromium                  gvfs-backends             libexif12                 libmousepad0              libunistring5             python3-bcrypt            ruby-oj
  chromium-common           gvfs-common               libexiv2-28               libmpg123-0t64            liburing2                 python3-brlapi            ruby-snmp
  chromium-sandbox          gvfs-daemons              libexiv2-data             libneon27t64              libvpl2                   python3-cherrypy3         ruby-spider
  clang                     gvfs-fuse                 libexpat1                 libneon27t64-gnutls       libwbclient0              python3-cli-helpers       ruby-typhoeus
  console-setup             gvfs-libs                 libexpat1-dev             libnetcdf22               libwebkit2gtk-4.1-0       python3-cryptography      ruby-zip
  console-setup-linux       gvmd                      libffado2                 libnfsidmap1              libwebsockets19t64        python3-django            runit-helper
  cpp-15                    gvmd-common               libfluidsynth3            libnghttp2-14             libwinpr3-3               python3-docx              samba
  cpp-15-x86-64-linux-gnu   ibus-gtk3                 libfreerdp-client3-3      libnghttp3-9              libx11-6                  python3-ephem             samba-common
  curl                      ibus-gtk4                 libfreerdp3-3             libngtcp2-16              libx11-data               python3-filelock          samba-common-bin
  dbus                      imagemagick-7-common      libfreetype6              libngtcp2-crypto-gnutls8  libx11-dev                python3-fonttools         samba-libs
  dbus-bin                  inetutils-telnet          libgcc-15-dev             libngtcp2-crypto-ossl0    libx11-xcb1               python3-gi                sane-airscan
  dbus-daemon               intel-microcode           libgcc-s1                 libnss3                   libxfce4panel-2.0-4       python3-gi-cairo          sleuthkit
  dbus-session-bus-common   iproute2                  libgcrypt20               libobjc-15-dev            libxml-parser-perl        python3-gssapi            smbclient
  dbus-system-bus-common    iptables                  libgdk-pixbuf-2.0-0       libobjc4                  libxmp4                   python3-gvm               socat
  dbus-user-session         ipython3                  libgdk-pixbuf2.0-bin      libout123-0t64            libxtables12              python3-ipython           speech-dispatcher
  dbus-x11                  kali-archive-keyring      libgdk-pixbuf2.0-common   libpaho-mqtt1.3           libze1                    python3-jinja2            speech-dispatcher-audio-plugins
  debconf                   kali-defaults             libgfortran5              libpam-gnome-keyring      llvm-spirv-18             python3-jwt               speech-dispatcher-espeak-ng
  debconf-i18n              kali-defaults-desktop     libgirepository-2.0-0     libpgm-5.3-0t64           mariadb-client            python3-kombu             sqlmap
  debugedit                 kali-desktop-base         libglib2.0-0t64           libphonenumber8           mariadb-client-core       python3-ldb               squashfs-tools
  dhcpcd-base               kali-desktop-core         libglib2.0-bin            libpipewire-0.3-common    mariadb-common            python3-linkify-it        tcpdump
  dirmngr                   kali-desktop-xfce         libglib2.0-data           libplacebo360             mariadb-plugin-provider-bzip2 python3-llvmlite      tdb-tools
  dmidecode                 kali-linux-core           libgnutls-dane0t64        libpq5                    mariadb-plugin-provider-lz4 python3-louis           tpm-udev
```

STEP 2 : Upgrade system including dependencies

```
┌──(aman㉿kali)-[~]
└─$ sudo apt full-upgrade
[sudo] password for aman:
The following packages were automatically installed and are no longer required:
  libann0  libcdt5  libcgraph6  libdlt2  libgdk-pixbuf2.0-bin  libicu76  liblab-gamut1  libsimdutf27  libtsk19t64  libtss2-fapi1t64  libzxing3
Use 'sudo apt autoremove' to remove them.

Upgrading:
  colord    libcolorhug2  libgbm1          libglx-mesa0             libpipewire-0.3-0t64     libspa-0.2-bluetooth  mesa-libgallium  pipewire      pipewire-pulse      python3-tables-lib
  graphviz  libegl-mesa0  libgl1-mesa-dri  libonnxruntime-providers libpipewire-0.3-modules  libspa-0.2-modules    nodejs           pipewire-bin  python3-pygraphviz

Installing dependencies:
  libada-url0-3  libblosc2-6  libgusb2a  libgvc7  libgvplugin-gd8  libgvplugin-neato-layout8  libgvplugin-pango8  libnode127  libonnx1l  libonnxruntime1.23  libprotobuf-lite32t64  libsimdjson30  libxdot4

REMOVING:
  libblosc2-4  libgusb2  libgvc6  libnode115  libonnx1t64  libonnxruntime1.21  mesa-va-drivers

Summary:
  Upgrading: 19, Installing: 13, Removing: 7, Not Upgrading: 0
  Download size: 39.0 MB
  Space needed: 19.7 MB / 12.9 GB available

Continue? [Y/n] y
Get:1 http://http.kali.org/kali kali-rolling/main amd64 python3-tables-lib amd64 3.10.2-3+b2 [707 kB]
Get:5 http://mirrors.esto.network/kali kali-rolling/main amd64 libgusb2a amd64 0.4.9-7 [38.2 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 libblosc2-6 amd64 2.22.0+ds-1 [202 kB]
Get:3 http://kali.download/kali kali-rolling/main amd64 libcolorhug2 amd64 1.4.8-3 [40.4 kB]
Get:4 http://kali.download/kali kali-rolling/main amd64 colord amd64 1.4.8-3 [205 kB]
Get:7 http://kali.download/kali kali-rolling/main amd64 libgvc7 amd64 14.1.2-1 [376 kB]
Get:6 http://mirrors.esto.network/kali kali-rolling/main amd64 libxdot4 amd64 14.1.2-1 [21.9 kB]
Get:15 http://http.kali.org/kali kali-rolling/main amd64 nodejs amd64 22.22.1+dfsg+~cs22.19.15-1 [373 kB]
Get:8 http://kali.download/kali kali-rolling/main amd64 graphviz amd64 14.1.2-1 [60.6 kB]
Get:9 http://kali.download/kali kali-rolling/main amd64 python3-pygraphviz amd64 1.14-5 [87.7 kB]
Get:10 http://kali.download/kali kali-rolling/main amd64 libgvplugin-gd8 amd64 14.1.2-1 [27.3 kB]
Get:11 http://kali.download/kali kali-rolling/main amd64 libgvplugin-pango8 amd64 14.1.2-1 [31.6 kB]
Get:12 http://kali.download/kali kali-rolling/main amd64 libgvplugin-neato-layout8 amd64 14.1.2-1 [291 kB]
Get:13 http://kali.download/kali kali-rolling/main amd64 libada-url0-3 amd64 3.4.3-1 [209 kB]
Get:14 http://http.kali.org/kali kali-rolling/main amd64 libsimdjson30 amd64 4.3.1+really4.3.1-1 [76.2 kB]
Get:16 http://http.kali.org/kali kali-rolling/main amd64 libnode127 amd64 22.22.1+dfsg+~cs22.19.15-1 [14.7 MB]
Get:18 http://mirrors.esto.network/kali kali-rolling/main amd64 libegl-mesa0 amd64 26.0.2-1 [133 kB]
Get:19 http://mirrors.esto.network/kali kali-rolling/main amd64 libgbm1 amd64 26.0.2-1 [54.6 kB]
Get:26 http://mirrors.esto.network/kali kali-rolling/main amd64 pipewire-pulse amd64 1.6.2-1 [15.1 kB]
Get:17 http://kali.download/kali kali-rolling/main amd64 libgl1-mesa-dri amd64 26.0.2-1 [43.4 kB]
Get:20 http://kali.download/kali kali-rolling/main amd64 libglx-mesa0 amd64 26.0.2-1 [125 kB]
Get:21 http://kali.download/kali kali-rolling/main amd64 mesa-libgallium amd64 26.0.2-1 [10.4 MB]
Get:22 http://kali.download/kali kali-rolling/main amd64 libprotobuf-lite32t64 amd64 3.21.12-15 [281 kB]
Get:23 http://kali.download/kali kali-rolling/main amd64 libonnx1l amd64 1.20.0-4 [1,382 kB]
Get:24 http://http.kali.org/kali kali-rolling/main amd64 libonnxruntime-providers amd64 1.23.2+dfsg-6+b2 [310 kB]
Get:25 http://http.kali.org/kali kali-rolling/main amd64 libonnxruntime1.23 amd64 1.23.2+dfsg-6+b2 [5,345 kB]
Get:27 http://kali.download/kali kali-rolling/main amd64 libpipewire-0.3-modules amd64 1.6.2-1 [839 kB]
Get:28 http://kali.download/kali kali-rolling/main amd64 libpipewire-0.3-0t64 amd64 1.6.2-1 [305 kB]
Get:29 http://kali.download/kali kali-rolling/main amd64 libspa-0.2-bluetooth amd64 1.6.2-1 [460 kB]
```

Key Observations

- Upgrading improves security and performance.
- Some upgrades modify dependencies.
- Regular updates prevent vulnerabilities.

What I learned

System updates are essential for maintaining security, stability, and compatibility of software.

Question 5

Select a software package from GitHub and clone it to the local system

Approach

I used Git to clone an open-source software repository from GitHub to my local system to understand how security tools are shared and installed manually.

Commands

- `git clone <repository-link>` → downloads repository from GitHub
- `cd <directory-name>` → move into cloned directory
- `ls` → list files inside repository

Output

STEP 1 : Clone repository from GitHub

```
┌──(aman㉿kali)-[~]
└─$ git clone https://github.com/sqlmapproject/sqlmap.git
Cloning into 'sqlmap'...
remote: Enumerating objects: 88784, done.
remote: Counting objects: 100% (7/7), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 88784 (delta 2), reused 1 (delta 1), pack-reused 88777 (from 2)
Receiving objects: 100% (88784/88784), 85.33 MiB | 897.00 KiB/s, done.
Resolving deltas: 100% (70378/70378), done.
```

STEP 2 : Move into cloned directory

```
┌──(aman㉿kali)-[~]
└─$ cd sqlmap
```

STEP 3 : List repository files

```
┌──(aman㉿kali)-[~/sqlmap]
└─$ ls
data  doc  extra  lib  LICENSE  plugins  README.md  sqlmapapi.py  sqlmapapi.yaml  sqlmap.conf  sqlmap.py  tamper  thirdparty
```

Key Observations

- GitHub provides open-source tools used in cybersecurity.
- Cloning downloads full source code to local system.

- Many security tools are installed manually using Git.

What I learned

GitHub is an important platform for accessing cybersecurity tools and learning from open-source projects.

Security Insights

- Security professionals must keep systems updated to patch vulnerabilities.
- Outdated packages may contain known exploits that attackers can use.
- GitHub repositories frequently host open-source security tools used in penetration testing.
- Attackers often install tools from repositories to perform scanning, exploitation, and enumeration.
- Dependency management is important because vulnerable libraries can compromise the entire system.